**MKT 3717 Product and Brand Management** by Dr. Dayoung Kim

# Final Test Study Guide

**Location: LT16 Time: 26 March (Thurs) 2026, 12:00pm-1:45pm**

For this test, you are responsible for all materials presented and discussed in class including the case/article discussions, and the practice problems. This test will consist of mix of multiple-choice questions, short-answer, and short-essay questions. It consists of 20% of your module grade. Test starts at the beginning of the class (at 12pm) and will be a closed-book and closed-notes test through Examsoft. You need to **bring your own laptop with *Examplify* (software) installed.** If you haven't downloaded this software on your laptop, please go to: https://nus.atlassian.net/wiki/spaces/DAstudent/pages/22511850/Registering+Examplify+on+your+laptop+Win+Mac and read 'Getting Started' instructions and watch the briefing video. This is a closed-book and closed-notes test. You may bring a **pen/pencil** and a personal **calculator**. I will give you the scratch paper, and Examsoft screen will have embedded calculator functions.

The most important study materials are your lecture notes. In any case where the textbook definition happens to be slightly different from lecture slides, follow the lecture slide definitions and feel free to ask me for clarifications. Below are the important points that we covered in this course. You can use the list to check your level of understanding of the class materials, but keep in mind that this list is not exhaustive. Although I won't ask questions from the textbook that is not discussed in class, Chapter 1: *The Strategic Elements of Product Development* and Chapter 2: *The New Products Process* of your textbook 'New Products Management' by Crawford textbook overlap with the first few lectures and provide good references on the new product development process.

1. New Product Development
- What is New Product? Why difficult?
2. Opportunity Identification
- How to understand the marketplace and identify opportunities (Situation Analysis)
    - RWW framework
    - General Environment Analysis (Political-Economic-Social-Technological)
    - Industry Environment Analysis (Porter's Five-force Model)
- Market sizing (simple exercises), Benefit of Jobs-to-be-done approach
- Generating Product Specification
    - Translating customer voice to customer needs statement
    - Translating needs to product specifications, considerations in freezing time
3. Concept Generation and Evaluation
- Product concept definition and inputs (Crawford p. 96-102)
- Concept selection methods (pros/cons)

    - Concept screening matrix
    - Concept scoring matrix
- Concept testing methods
    - Conjoint analysis: constructing variables, estimate using regression approach, and interpretation of results
    - Use of conjoint analysis (pros/cons) (Crawford p. 147-151)

4. Product Design and Architecture

- Modular design for mass customization

5. Market Testing Methods – Experimental approaches

- Why building causality is important, and how
    - A/B testing
    - Factorial Design
    - Basic interpretation of results (confidence interval, p-value, and regression)

6. Project Management

- Characteristics of Project
- Using Network Diagram for Project Management
    - Work-breakdown-structure, Identify critical path (Practice problems)
    - Slack time calculation, Deciding project crashing

7. Brand Management Overview

- Building blocks of brand (definitions)
- Keller's brand resonance pyramid
- Modern brand management challenges (e.g., sustaining continuity vs. staying relevant)

## Example Multiple Choice Questions:

1. Ace Corporation is a manufacturer of powder and liquid dishwashing detergents. Owing to popular demand, the company introduces a single load, scented, gel detergent encased in water-soluble bags designed to fit dishwasher detergent trays. The firm's new product is best described as a(n):

A. new-to-the-world product

B. line extension

C. repositioned product

D. improvised product

E. cost reduction

Answer: (B) Product line extensions are additions to existing product lines of a firm, which are designed to flesh out the product line as offered to the firm's current markets.

2. Which one of the following best describes the critical path of a PERT network for project management?

A. the sequence of activities between a project's start and finish that takes the longest time to complete

B. the sequence of activities between a project's start and finish that has the maximum amount of activity slack

C. the set of activities that has no precedence relationships

D. the sequence of activities that has the lowest normal activity cost

E. None of the above

Answer: (A)

## Example Short Answer and Essay Questions:

1. What are the key benefits of designing the product to be modular under demand uncertainty? Describe within 3-4 sentences.

For question 2 and 3, refer to a customer survey result table below. A consumer ranked her preference on the combination of three attributes of a smartphone (i.e., smaller number in rank means higher preference). For each attribute, there are only two levels: small or wide for screen size, 10hr or 16hr for battery life, and iPhone and Samsung for brand. (Hint: Recall how the average ranking differences match with the regression coefficients of conjoint analysis).

| Combination | | | Consumer's |
|---|---|---|---|
| Screen size | Battery Life | Brand | Rank |
| small | 10hr | iphone | 8 |
| wide | 16hr | iphone | 3 |
| wide | 10hr | samsung | 2 |
| small | 16hr | iphone | 7 |
| small | 10hr | samsung | 6 |
| wide | 16hr | samsung | 1 |
| wide | 10hr | iphone | 4 |
| small | 16hr | samsung | 5 |

2. Which attribute is considered the most importantly for this consumer?

Your answer: _____

3. For this consumer, distribute the relative importance of the three attributes.

Your answer: Screen size _____%

Battery life _____%

Brand _____%

(Total sum to 100%)